

Bernice Animwaa Boateng

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SUMMARY

Software engineer and data analyst with a computer engineering degree and hands-on machine learning experience in healthcare. At Really Great Tech I work on ML pipelines for clinical signal data and an AI system that detects distress in phone calls. On my own I build and deploy full-stack ML apps, from training the model to shipping a live API. Looking for a junior software, data or AI role.

EXPERIENCE

Software Engineer and Data Analyst

Dec. 2025 – Present

Really Great Tech

Accra, Ghana

SEMLER: Worked with senior data analysts on a model that predicts myocardial strain from PPG signals, using XGBoost regression and Leave-One-Clinic-Out validation so the model is always tested on clinics it has never seen. Built feature extraction and recursive feature elimination steps, tuned loss functions, and wrote pipeline scripts for leakage-safe hyperparameter tuning.

AppWork VoiceAI: Helped build an AI system for a property management company that flags distress in inbound calls. Labelled 911 call data and built a pipeline that combines audio and text signals.

Software Engineer and IT Intern

Apr. 2025 – Aug. 2025

Cocoa Clinic

Accra, Ghana

Helped maintain and update the hospital's electronic health record (EHR) systems.

Worked with the IT team on digital transformation projects and internal reporting tools.

Handled technical support for internal software and hardware issues.

PROJECTS

Vellichor AI – Emotion & Theme Analysis | [FastAPI](#), [GPT-2](#), [Hugging Face](#), [Docker](#) [Live App](#) | [Backend](#) | [ML Model](#)

Fine-tuned GPT-2 end to end to classify the emotion and subtext of written dialogue into 7 categories, served through a FastAPI backend.

Used context-aware features and dropout regularization to read subtext instead of surface keywords, and it outperformed standard models on nuanced fiction.

MedMate DDI – Drug Interaction Checker | [Python](#), [React](#), [Django](#), [Transformers](#), [XGBoost](#) [Live App](#) | [ML Model](#)

Co-developed a full-stack app for a three-person final year project that detects drug-drug interactions, using DrugBank and PubChem data.

Built the severity classification models, XGBoost and Random Forest, reaching 0.89 and 0.86 training accuracy, and led the transformer model that generates interaction descriptions, reaching a BLEU score of 0.96.

Deployed the Django REST API on Render and the React frontend on Vercel.

CardioGlobal – Heart Disease Data Visualization | [Python](#), [Dash](#), [Plotly](#), [scikit-learn](#), [LightGBM](#) [Live App](#)

Built an interactive dashboard for clinicians and researchers that combines WHO/GHO global mortality data with the UCI Cleveland Heart Disease dataset.

Added an ML Insights tab with a Random Forest risk predictor (81.5% accuracy, 0.90 ROC-AUC), a LightGBM mortality forecaster (R^2 0.78 on held-out years), and an HDBSCAN country clustering model.

Book App & Reader Community Platform | [React](#), [Backend APIs](#), [Database Design](#)

Building a book-centric application that allows users to discover books and connect within a reader-focused social community

Designing features for user interaction, social engagement, and content sharing to foster meaningful connections among readers

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS, SQL

AI/ML: TensorFlow, Hugging Face Transformers, XGBoost, LightGBM, Random Forest, GPT-2 fine-tuning

Frameworks & Databases: React.js, Node.js, Flask, Django, FastAPI, MongoDB, PostgreSQL, Supabase

Tools: Render, Vercel, Git, Docker, Postman, Figma, Microsoft Word, Microsoft Excel

EDUCATION

University of Ghana – B.S. Computer Engineering

Accra, Ghana

Sep. 2021 – Nov. 2025